IMO Training – Number Theory
Law of Quadratic Reciprocity
Miscellaneous Facts in Number Theory

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Introduction

Unless otherwise specified, we shall focus on integers. Furthermore, let p be an odd prime.

Definition. Let $(a,m)=1$. Then a is a *quadratic residue mod* m if $\exists\, x \in \mathbb{Z}$ such that $x^2 \equiv a \pmod{m}$. Otherwise, we call a is a *quadratic nonresidue mod* m.

Definition. The *Legendre's Symbol* $\left(\dfrac{a}{p}\right)$ is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} +1 & \text{if } a \text{ is a quadratic residue mod } p \\ -1 & \text{if } a \text{ is a quadratic nonresidue mod } p \\ 0 & \text{if } p \mid a \end{cases}$$

Theorem. (Quadratic Reciprocity Law) Let p, q are distinct odd primes, then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)}.$$

Theorem. (Fermat's Two Square Theorem) Any prime p of the form $4k+1$ is a sum of two squares.

Theorem. (Fermat's Last Theorem) If $x, y, z, n \in \mathbb{Z}$, $n \geq 3$ and $x^n + y^n = z^n$, then $xyz = 0$.

Theorem. (Dirichlet) If $(a,b)=1$, then there are infinitely many primes of the form $an+b$.

Theorem. (Catalan) The only integral solution to the equation $a^b - c^d = 1$ with $a, b, c, d > 1$ is $(a,b,c,d) = (3,2,2,3)$.

Theorem. (Fibonacci Power) 1,8, and 144 are the only perfect powers in the Fibonacci sequence.

Proposition. (Euler's Criterion) For all a, $\left(\dfrac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$.

Proposition. $\left(\dfrac{ab}{p}\right) = \left(\dfrac{a}{p}\right)\left(\dfrac{b}{p}\right)$

Example. Find $\left(\dfrac{-1}{p}\right)$ and $\left(\dfrac{2}{p}\right)$.

Example. Evaluate the following:

Example. Prove that -3 is not a primitive root of any prime of the form $4^n + 3$.

Example. Solve $x^2 + 3xy - 2y^2 = 122$ for integers.

Example. Show that there are infinitely many primes of the form $4k + 1$.

Example. Show that $a^2 + b$ and $b^2 + a$ cannot be both perfect squares.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $x^5 = y^2 + 4$.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $3x^2 + 1 = 4y^3$.

Exercises

1. Find all $x, y \in \mathbb{Z}$ s.t. $x^3 - y^3 = xy + 61$.
2. Find all $x, y \in \mathbb{Z}$ s.t. $x^2 - y! = 2001$.
3. For $m, n \in \mathbb{N}$, find the minimum value of $|12^m - 5^n|$.
4. Find all $x, y \in \mathbb{Z}$ s.t. $x^3 + y^4 = 19^{19}$.
5. Find all $k \in \mathbb{N}$ s.t. $\exists\ a, b \in \mathbb{N}$ with $a^2 + b^2 = kab$.
6. If $a, b \in \mathbb{N}$ and $b^2 + ab + 1 \mid a^2 + ab + 1$, prove that $a = b$.
7. Show that for $a, b, c \in \mathbb{N}$ with $a^2 + b^2 = c^2$, $\exists\ m, n \in \mathbb{Z}$ s.t. $(a, b, c) = (m^2 - n^2, 2mn, m^2 + n^2)$.
8. Find all $x, y \in \mathbb{Z}$ s.t. $x^2 - 2y^2 = 1$.
9. (Bulgaria 1998) Let $m, n \in \mathbb{N}$ such that $A = \dfrac{(m+3)^n + 1}{3m} \in \mathbb{Z}$. Prove that A is odd.
10. (IMO 1996) Let $a, b \in \mathbb{N}$ s.t. $15a + 16b, 16a - 15b$ are both perfect squares. Find the minimum value of the smaller square.
11. (Bulgaria 1996) Find all primes p, q such that $pq \mid (5^p - 2^p)(5^q - 2^q)$.
12. (USA TST 2003) Find all ordered primes p, q, r s.t. $p \mid q^r + 1, q \mid r^p + 1, r \mid p^q + 1$.
13. (China 2009) Find all primes p, q s.t. $pq \mid 5^p + 5^q$.

Extra Space